

Term	Definition
Qualification	Operation to demonstrate that an analytical system or instrument is functioning properly and yielding the expected results
Quality Control	Sample adapted to the method used and intended to assess the accuracy and precision of the results
Quantity	Property of a phenomenon, body, or substance, where the property has a magnitude that can be expressed as a number and a reference
Random error	Component of measurement error that, in replicate measurements, varies in an unpredictable manner
Range	Range of the interval $[a, b]$ is the difference $b - a$, and is denoted by $r[a, b]$
Recovery yield	Percentage of the known or assigned concentration of an analyte recovered thanks to the analytical procedure
Reference (or official) measurement procedure	Measurement procedure accepted as providing measurement results fit for their intended use in assessing measurement trueness
Reference material (RM)	Material, sufficiently homogeneous and stable with reference to specified properties, that has been established to be fit for its intended use in measurement or in examination of nominal properties
Reference value	Value used as a basis for comparison for values of the same nature
Rejection interval	Interval of non-permissible measured values
Relative standard uncertainty	Standard measurement uncertainty divided by the absolute value of the measured value
Repeatability	Measurement precision under a set of repeatability conditions of measurement
Repeatability condition	Condition of measurement, out of a set of conditions that includes the same measurement procedure, same operators, same measuring system, same operating conditions, and same location, and replicate measurements on the same or similar objects over a brief period
Reproducibility	Measurement precision under a set of reproducibility conditions of measurement
Reproducibility condition	Out of a set of conditions that includes various locations, operators, measuring systems, and replicate measurements on the same or similar objects
Sampling	Act permitting the obtaining of a biological sample or entity taken or, by extension, the result of that act
Sensitivity	Quotient of the change in an indication of a measuring system and the corresponding change in a value being measured
Series	Set of measurements, replicated or not, performed under repeatability condition

Term	Definition
Specification Specified requirement	Need or expectation that is stated. Specified requirements may be stated in normative documents such as regulations, standards, and technical specifications. A typical specified requirement takes the form of a stated interval of permissible values of a measurable property of an item
Specification limit	Specified upper or lower bound of permissible values of a property
Specificity	Capability of a measuring system or operating procedure to measure the concentration of a given analyte
Standard addition or Spike	Addition to a material or standard of chemical composition defined with uncertainty in a known quantity. Conventionally, this operation is intended to confirm the trueness of a method or to calibrate an instrument
Standard uncertainty	Measurement uncertainty expressed as a standard deviation
Statistical dispersion interval	Interval determined from a random sample so that there is a specified level of confidence that the interval covers at least a given proportion of the sampled population
Systematic error	Component of measurement error that in replicate measurements remains constant or varies in a predictable manner
Target uncertainty	Measurement uncertainty specified as an upper limit and decided based on the intended use of measurement results
Tolerance interval (conformity assessment)	Interval of permissible values of a property. Unless otherwise stated in a specification, the tolerance limits belong to the tolerance interval. This term used in conformity assessment has a different meaning than as used in statistics
True value	Value consistent with the definition of a quantity
Trueness	Closeness of agreement between the average of an infinite number of replicate measures of a value and a reference value
Type A evaluation	Evaluation of a component of measurement uncertainty by a statistical analysis of measured values obtained under defined measurement conditions
Type B evaluation	Evaluation of a component of measurement uncertainty determined by means other than type A evaluation
Uncertainty budget	Statement of a measurement uncertainty, of the components of that measurement uncertainty, and of their calculation and combination
Validation	Verification, where the specified requirements are adequate for an intended use
Validation criterion	Element to which one refers to judge, assess, or define whether a method is valid

Term	Definition
Validation range	Concentration range and matrix types covered by the validation study
Validation standard	Material selected as representative of the matrix to which the validation study relates. It must be homogenized and of known concentration. In the most favorable case, it is a RM sometimes certified CRM
Verification	Provision of objective evidence that a given item fulfills specified requirements
Within-series variance	Refers to the influence of replicate measures on identifiable groups (related to ANOVA)
Working interval	Set of values of quantities of the same kind that can be measured by a given measuring instrument or measuring system with specified instrumental measurement uncertainty, under defined conditions
β -Expectation Tolerance interval	Statistical dispersion interval, with an expected coverage proportion $\beta\%$ of possible values
β - γ -Content Tolerance interval	Statistical dispersion interval, with a coverage proportion $\beta\%$ of values, with a confidence level $\gamma\%$

Acronyms

Acronym	Expanded name
A2LA	American Association for Laboratory Accreditation
AFNOR	Association Française de Normalisation
AIDS	Acquired immunodeficiency syndrome
ANOVA	Analysis of variance
AOAC	Association of Official Analytical Chemists
API	Active principal ingredient
BIPM	Bureau International des Poids et Mesures
CAC	Commission of the Codex Alimentarius
CCMAS	Codex Committee on Methods of Analysis and Sampling
CCQM	Consultative Committee on the Quantity of Matter
CI	Confidence interval
CIPM	Convention Internationale des Poids et Mesures
COFRAC	Comité Français d'Accréditation
COMAR	Code of reference materials
CRM	Certified reference material
df	Degrees of freedom
EC	External calibration

Acronym	Expanded name
EDQM	European Directorate for the Quality of Medicines
EMA	European Medicines Agency
EPA	U.S. Environmental Protection Agency
ERM	External reference material
FDA	Food and Drug Administration
GEON	OMCL (official medicines control laboratory) network of the Council of Europe
GUM	Guide to the expression of uncertainty in measurement
HPSAM	H-point standard addition method
HR-MS	High-resolution mass spectrometry
IC	Internal calibration
ICH	Before 2015: International Conference on Harmonization of technical requirements for registration of pharmaceuticals for human use
ICH	International Council for Harmonization of technical requirements for registration of pharmaceuticals for human use
ICP-ID-MS	Isotope dilution inductively coupled plasma mass spectrometry
IEC	International Electrotechnical Commission
IFCC	International Federation of Clinical Chemistry and Laboratory Medicine
ILAC	International Laboratory Accreditation Cooperation
IPD	Isotopic pattern deconvolution
IRM	Internal reference material
IS	Internal standard
ISC	In-sample calibration
ISO	International Organization for Standardization
IUPAC	International Union of Pure and Applied Chemistry
IUPAP	International Union of Pure and Applied Physics
JCGM	Joint Committee for Guides on Metrology
LC	Liquid chromatography
LC-MS	Liquid chromatography–mass spectrometry
LC-MS/MS	Liquid chromatography coupled to tandem mass spectrometry
LD ₅₀	Level of detection at 50%
LIMS	Laboratory information management system
LLOQ	Lower limit of quantification
LOD	Limit of detection
LOQ	Limit of quantification
MAP	Method accuracy profile
MIRM	Multiple isotopologue reaction monitoring
MRL	Maximum residue limit

Acronym	Expanded name
MS	Mass spectrometry
MU	Measurement uncertainty
NIRS	Near-infrared spectrometry
OIML	Organisation Internationale de Métrologie Légale
OLS	Ordinary least-squares
OMCL	Official medicines control laboratory
OOS	Out of specification
QC	Quality control
RF	Response factor
RM	Reference material
<i>RSD</i>	Relative standard deviation or coefficient of variation
SAM	Standard addition method
SEC	Exclusion chromatography or molecular sieve chromatography
SFSTP	Société Française des Sciences et Techniques Pharmaceutiques
SI	International System (of units)
SIL	Stable isotope labeled
SMPR	Standard method performance requirement
SPC	Statistical process control
std. dev.	Standard deviation
TDI	Total daily intake
TI	Tolerance interval
TRV	Toxicological reference value
<i>ULOQ</i>	Upper limit of quantification
USP	US pharmacopeia
UV	Ultraviolet
VIM	International Vocabulary of Metrology
WADA	World Anti-doping Agency
WHO	World Health Organization
WLS	Weighted least-squares
β -ETI	β -Expectation tolerance interval
β - γ -CTI	β - γ -Content tolerance interval

Reference

- 1 BIPM, IEC, IFCC, ILAC, ISO, IUPAC, IUPAP, and OIML (2012). *International Vocabulary of Metrology — Basic and General Concepts and Associated Terms (VIM3)*. Sèvres, France: JCGM 200, <https://www.bipm.org/> (accessed 23 July 2023).